CINZIA

AI / Machine Learning Engineer – Applied AI

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PROFESSIONAL EXPERIENCE

AI Systems Engineer — Framatome, DTI

La Défense, France

Primary project: End-to-end development of an industrial computer vision system for real-time zirconium defect detection

- **Data & labeling:** Labeled the full industrial dataset with Label Studio; prepared and analyzed image data, investigated challenging samples and contributed to preprocessing and data augmentation strategies.
- **Deep Learning & Computer Vision:** Developed and trained CNNs and YOLOv8 object detection models for industrial visual inspection; evaluated with precision, recall and mAP and iteratively improved the models.
- **Deployment:** Integrated the vision model into an embedded iPad application for real-time defect detection with the device camera, a decision-support tool used by field experts at the end of the production line.
- **Industrialization:** Built computer vision pipelines under real industrial constraints (data quality, robustness, latency, production deployment), in close collaboration with operational experts.
- **Generative AI R&D:** Studied and implemented diffusion models to synthesize realistic defects for data augmentation, using JAX for optimization and differentiable computation.
- **Deep Learning & LLM research:** Conducted R&D on Transformers, Large Language Models and embeddings, with weekly in-depth technical presentations on the theory and applied mathematics behind deep learning and generative architectures (e.g. DALL·E).

Technical Environment: Python, PyTorch, JAX, CNN, YOLOv8, Diffusion Models, Transformers, LLMs, Embeddings, Label Studio, NVIDIA V100, SLURM, Docker, Linux, Git

ACADEMIC PROJECTS

- **Bike Rental Demand Forecasting (London)** — Hourly demand over 2 years (~17k obs.); calendar/weather features, ACF/PACF and stationarity analysis; benchmarked GLM, SARIMA, GAM, Random Forest, XGBoost, MLP and Prophet. Best: XGBoost (RMSE ≈ 373), plus EWA model aggregation.
- **Online Convex Optimization & SVRG** — Implemented GD, SGD, Projected GD, AdaGrad, Mirror Descent, Online Newton Step and SVRG (Johnson & Zhang, NeurIPS 2013); convergence analysis; applied to linear SVM on high-dimensional image data.
- **Multi-Model Vehicle Classification** — End-to-end pipeline (ANOVA, PCA); benchmarked 10 models from Logistic Regression to MLP and ensembles, with GridSearchCV, 10-fold cross-validation and bias–variance analysis.
- **Parallel Computing — Conway's Game of Life** — Parallel cellular automaton simulation; performance and scalability benchmarking.
- **Generative Models — Research Review** — Critical analysis and presentation of a research article on generative modeling.

EDUCATION

Double Master's Degree – Statistical Engineering & Data Science

ISUP & Sorbonne University

Paris, France | 2022 – 2024

- **Key coursework:** Machine Learning & Deep Learning, Advanced Statistics, Probability & Optimization, Computer Vision

PROFILE

Machine Learning Engineer focused on building and industrializing robust AI systems from experimentation to production.

Strong background in applied mathematics, optimization, statistical modeling, and deep learning, with hands-on experience in computer vision, generative AI, and real-time ML systems.

Experienced in translating complex technical and operational constraints into scalable, reliable AI solutions, with a strong focus on measurable performance and real-world impact.

Interested in high-impact AI applications across healthcare, security, finance, and other decision systems.

SKILLS

Programming & Data

Python | C++ | SQL | Bash | NumPy | Pandas | Scikit-learn

Machine Learning

PyTorch | Computer Vision | Deep Learning | YOLOv8 | Statistical Learning | Time Series | Representation Learning

Generative & Foundation Models

LLMs | Diffusion Models | Vision Transformers | Fine-tuning | Embeddings

Cloud, MLOps & Compute

Azure | GCP | MLflow | Docker | Git | Linux | CI/CD | CUDA | JAX | NVIDIA V100 | SLURM | Label Studio | Model Deployment

LANGUAGES

- French: Native
- English: C2
- Spanish: B2

RESEARCH INTERESTS

- Foundation Models & Generative AI
- Model Reasoning & Interpretability
- Robustness & Evaluation
- AI Agents